Bioresource Technology 393 (2024) 130092

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Performance and mechanism of various microaerobic pretreatments on anaerobic digestion of tobacco straw

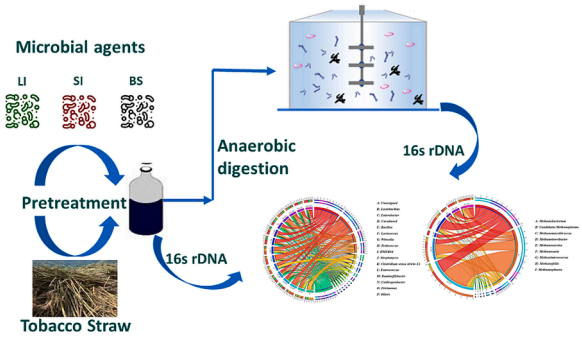
Ligong Wang, Wenxiong Jin, Fanfan Cai, Chao Song, Yan Jin, Guangqing Liu, Chang Chen *

College of Chemical Engineering, Beijing University of Chemical Technology, Beijing 100029, China

HIGHLIGHTS

- Effects of various microaerobic pretreatments on AD of tobacco straw were studied.
- Biogas slurry as pretreatment agent dosing 4 mL/g VS of O_2 (BS-4) performed better.
- BS-4 had a good performance in the lignocellulosic decomposition of tobacco straw.
- O_2 dosing in microbial pretreatment process could change the microbial diversity.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Oxygen concentration
Microbial pretreatment
Methane yield
Microbial community

ABSTRACT

Tobacco straw is an abundant biomass in China's agricultural ecosystems, and has high potential for methane production. However, the anaerobic digestion (AD) efficiency is limited by the recalcitrant lignocellulose structure of the tobacco straw. In this study, three microaerobic pretreatments were performed for the AD of tobacco straw to increase methane production. Among them, microbial pretreatment with biogas slurry at an oxygen concentration of 4 mL/g VS resulted in the highest methane production of 349.1 mL/g VS, increasing by 19.8 % than that of untreated. During this pretreatment, the relative abundances of *Enterococcus* and *Clostridium sensu stricto 12*, which are closely related to acetic acid production and cellulose degradation, were high, and these bacteria might have an important contribution to substrate hydrolysis and the methanogenesis efficiency of the AD process. This study advances the understanding of microaerobic pretreatment processes and provides technological guidance for the efficient utilization of tobacco straw.

1. Introduction

Tobacco plays an important role in China's economy, due to the first planting area and output worldwide. Based on the Statista data, the estimated tobacco production in 2022 is approximately 2.19 million metric tons. Zhang et al. (2023) had reported that tobacco straw accounts for more than 48 % of tobacco production. As a result, over 1.05 million metric tons of tobacco straw are generated a year and are either

* Corresponding author at: 505 A Zonghe Building, Beijing University of Chemical Technology, 15 North 3rd Ring East Road, Beijing 100029, China.
E-mail address: chenchang@mail.buct.edu.cn (C. Chen).

https://doi.org/10.1016/j.biortech.2023.130092
Received 25 September 2023; Received in revised form 21 November 2023; Accepted 21 November 2023
Available online 23 November 2023
0960-8524/© 2023 Elsevier Ltd. All rights reserved.

abandoned or burned in fields, which not only results in a wastage of biomass resources but also leads to severe environment pollution. On the other hand, the tobacco mosaic virus and nicotine in tobacco straw make straw return and composting greatly challenging, as these components may migrate into the soil to decrease the yield and quality of tobacco for the next season. Therefore, a disposal measure that is effective and eco-friendly must be developed.

Anaerobic digestion (AD) refers to the process in which anaerobic and facultative microorganisms transform biodegradable organics into methane and carbon dioxide through a series of biochemical reactions under oxygen-free conditions. Due to its low cost and environmental advantages, AD has frequently served as a method to recycle agricultural wastes. Like conventional agricultural residues, tobacco straw is produced in large quantities and is abundant in lignocellulose components. Therefore, tobacco straw may function as an effective supplement to existing feedstocks, but the recalcitrance of lignocellulosic structure presents a bottleneck for the utilization of the available organic matter in tobacco straw.

Pretreatment strategies, such as physical and chemical pretreatments, are indicated measures to reduce the naturally recalcitrant structure of lignocellulosic biomass and increase hydrolysis rate. While a good consequence is received, the investment of physical and chemical pretreatments on the equipment and chemical reagent is relatively high, which may bring significant difficulties for commercial practice (Chen et al., 2022). Recently, microaerobic pretreatment achieved by introducing a small amount of oxygen or air into a pretreatment reactor before AD has gradually attracted increasing attention due to its ease of operation, low cost, and potential effectiveness (Huiliñir et al., 2023). To the best of our knowledge, good AD performance after microaerobic pretreatment has been established for corn straw (Zhen et al., 2021), giant grass (Wang et al., 2021), rice straw (Amin et al., 2021b), and so on, but negative outcomes also exist (Díaz et al., 2011; Lim and Wang, 2013; Okuda et al., 2007). The effect of microaerobic pretreatment varies with feedstock, and little research has been conducted to provide insights with respect to the pretreatment of tobacco straw. Considering the uniqueness of tobacco straw, gaps related to the impacts of microaerobic pretreatment on AD should be addressed for the effective disposal and renewable energy generation of tobacco straw.

Overall, the aims of this study are to (1) offer cost-effective pretreatment strategies using various microbial consortia to increase methane production from tobacco straw, (2) identify the effects of different microaerobic pretreatments on the concentration variations in volatile fatty acids (VFAs), the kinetics of methane generation, and lignocellulosic structural alterations, and (3) evaluate the mechanisms of microbial responses to oxygen from the pretreatment process to the AD stage.

2. Materials and methods

2.1. Feedstocks and inoculum

Tobacco straw was obtained from a farm in Hebei Province. A chopper was used to process the straw to a length of approximately 1 cm, and a cutting mill was employed to reduce its size to approximately 0.5 mm prior to usage. Liquid inoculant (LI) and straw composting inoculant (SI) were commercially available for straw composting. The biogas slurry (BS) and inoculum were taken from a biogas plant in Beijing and subsequently digested for 2 weeks at room temperature to eliminate residual fermentable organic substances and release background methane.

2.2. Microaerobic pretreatment and anaerobic digestion

The microaerobic pretreatment experiment was carried out in 500 mL serum bottles with the working volume of 150 mL. First, 1 g of LI, SI, and BS was added to the reactor to pretreat 5 g of tobacco straw (on a volatile solids (VS) basis), and subsequently, deionized water was added until a working volume of 150 mL was reached. Then, the pretreatment reactors were flushed with high-purity nitrogen for 3–4 min and carefully sealed to remove background air. Afterwards, different amounts (0, 4, 8, 12, and 16 mL/g VS) of oxygen were introduced into the respective pretreatment groups to create the different microaerobic environments, and this atmosphere was maintained for 2 days at 37 °C. To simplify sample labeling, the sample microbially pretreated with SI at the oxygen concentration of 0 mL/g VS was labelled SI-0, and the names of the other groups were abbreviated in a similar way. The pretreatment experiment was performed in triplicate, and the reactors were manually mixed every 4 h for uniform blending.

AD was carried out after oxygen was completely consumed in each reactor. The organic loading and feed-to-inoculum ratio were 20 g VS/L and 1:1, respectively, and the working volume for AD was altered to 250 mL. Thus, the inoculum was added for the AD reactor until the amount in each reactor reached 20 g VS/L, and then deionized water was added according to the working volume of 250 mL. The control groups contained the pretreatment agent and inoculum were set to exclude methane production from the microbial agent, and untreated groups containing tobacco straw, inoculum, and the pretreatment agent were also prepared for comparison. Each group was tested in duplicate. Before the AD run, the pH in all reactors was adjusted to 7.5, the reactors were flushed with nitrogen to create an anaerobic environment, and then they were subjected to 50 days of AD at a temperature of 37 °C. Shaking was performed twice daily by hand to ensure uniform mixing. The headspace pressure was determined using a 3151 WAL-BMP-Test system pressure gauge (WAL Messund Regelsysteme GmbH, Oldenburg, Lower Saxony, Germany) to compute the biogas yield. Biogas volume was determined by Eq. (1) (Amin et al., 2021b):

$$V_{biogas} = \frac{\Delta P \times V_{head} \times C}{R \times T} \tag{1}$$

where V_{biogas} is the daily biogas yield (L), ΔP represents the absolute pressure difference inside the reactor before and after the discharge of the biogas under water (KPa), V_{head} is the volume of headspace (L), C denotes the molar volume (22.41 L/mol), T indicates the absolute temperature (K), and R reflects the universal gas constant (83.14 L·mbar/K/mol). The methane content of biogas and oxygen concentration of the headspace in the pretreatment reactor were determined daily by a gas chromatograph (Agilent 7890B, Santa Clara, CA, USA), equipped with an analytical column (Agilent Hayesep Q) and a thermal conductivity detector.

The theoretical maximal methane production (*MMP*) of tobacco straw was calculated at an absolute temperature of 273.15 K and a pressure of 1013.25 mbar based on the Buswell method, as shown in Eqs. (2) and (3) and biodegradability (*BD*) was computed using Eq.(4) (Buswell and Mueller, 1952):

$$\begin{aligned} C_nH_aO_bN_c + (n - \frac{a}{4} - \frac{b}{2} + \frac{3c}{4})H_2O \rightarrow (\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8})CH_4 \\ + (\frac{n}{2} - \frac{a}{8} + \frac{b}{4} + \frac{3c}{8})CO_2 + cNH_3 \end{aligned} \tag{2}$$

$$MMP(mL/g\ VS) = \frac{22.4 \times 1000 \times (\frac{n}{2} + \frac{a}{8} - \frac{b}{4} - \frac{3c}{8})}{12n + a + 16b + 14c} \tag{3}$$

$$BD = \frac{EMY}{MMP} \times 100\% \tag{4}$$

where *EMY* is the highest cumulative methane yield, *MMP* refers to the theoretical maximal methane production from tobacco straw, and n, a, b, and c are stoichiometric coefficients.

2.3. Analysis methods

The total solids (TS) and VS of tobacco straw, microbial agent, and

inoculum were tested using the APHA method, 23rd edition (APHA, 2012). Microelements, including C, H, N, and S, were determined with a Vario EL CUBE element analyser (Germany). The content of O was computed using the formula C + H + O + N = 99.5 % on a VS basis. The determination of cellulose, hemicellulose, and lignin was based on the Van Soest method (Van Soest et al., 1991). The milled tobacco straw was determined for its lignocellulosic component, which includes neutral detergent fiber (NDF), acid detergent fiber (ADF) and acid detergent lignin (ADL). NDF and ADF determinations were fulfilled utilizing an ANKOM A2000i automatic fiber analyzer (ANKOM, USA) with filter bags. Sulfuric acid at a concentration of 72 % was added to digest the ADF, and the resulted residue is used to detect the ADL by the muffle furnace. Cellulose, hemicellulose, and lignin content was calculated based on the interrelationship among NDF, ADF, and ADL (Surendra et al., 2018). The sampled digestate was centrifuged at 10000 rpm for 10 min and then filtered using 0.45 μm and 0.22 μm filtration membranes in sequence, and the resultant supernatant was used to test the concentration and composition of VFAs. DNA of the bacterial and archaeal genomes was separated from the digestate using the OMEGA kit. 338F (ACTCCTACGGGAGGCAGCA) and 806R (GGACTACH VGGGTWTCTAAT) were used as bacterial primers to amplify in the V3-V4 gene region, while 515F (GTGCCAGCMGCCGCGGTAA) and 907R (CCGICAATTCMTTTRAGTTT) were used as archaeal primers to amplify in the V4-V5 gene region. The high-quality sequences for clustering as operational taxonomic units (OTUs) were obtained after removing chimeras and non-specifically amplified sequences, based on the lowest similarity level of 97 %. All data were processed using Origin, Circos software, as well as one-way and two-way analysis of variance with a statistical significance threshold of 0.05.

2.4. Kinetic analysis and microstructure analysis

The modified Gompertz model was utilized to the accumulated methane production data using Eq. (5).

$$Y = Y_0 exp\left\{ -exp\left[\frac{\mu_m e(\lambda - t)}{Y_0} + 1\right]\right\} \quad (5)$$

where Y is the simulated cumulative methane production (mL/g VS), Y_0 is the simulated maximum methane yield (mL/g VS), μ_m is the maximum methane production rate (mL/g VS/day), e is a constant with the value of 2.72, λ is the lag period (day), and t is the digestion time (day). The microstructure changes were tested using Fourier transform infrared (FTIR) spectroscopy and X-ray diffraction (XRD). The FTIR (Thermo Fisher Nicolet 6700, USA) is equipped with a DLATGS detector operating at scanning wavelengths in the range of 400 cm^{-1} to 4000 cm^{-1}, and the XRD (Bruker, Germany) is equipped with a Cu Kα target at a scanning rate of 5.0°/min ranging from 5° to 90°.

3. Results and discussion

3.1. Methane production

The characteristics of the substrates, inoculum, and microbial agents used are listed in Table 1, and according to the elemental composition, the *MMP* of tobacco straw was calculated to be 563.2 mL/g VS, which indicated that tobacco straw is a potential substrate for methane production. The cumulative methane yield from AD of tobacco straw after the application of different microaerobic pretreatments is presented in Fig. 1. The cumulative methane yields of SI-0, SI-4, SI-8, SI-12, and SI-16 were 315.9, 321.1, 257.7, 318.2, and 335.8 mL/g VS, respectively. The methane productions of the pretreated groups, except for the SI-8 group, were higher than that (298.7 mL/g VS) of the untreated tobacco straw, and the methane production of the SI-16 group was the highest, with an increase of 12.4 % compared with that of the untreated and other groups ($^*P < 0.05$). From the point of view of *MMP*, this observation revealed that the SI-16 pretreatment condition might have a better capacity to degrade tobacco straw. Interestingly, the SI-8 group resulted in the lowest methane production, and there was no significant difference between the SI-4 and SI-12 groups ($^*P > 0.05$). The decrease in methane yield in the SI-8 group compared to the untreated group might have occurred due to acid inhibition, and oxygen injected at concentrations of 4 and 12 mL/g VS possibly had a similar hydrolysis effect on tobacco straw. As shown in Fig. 1, the methane production of the LI-pretreated tobacco straw fluctuated from 252.4 to 293.3 mL/g VS with varying oxygen concentration. Clearly, the cumulative methane yield of the LI-16 group was the highest, with a 5.8 % enhancement compared with that (277.3 mL/g VS) of the untreated group ($^*P < 0.05$). After microaerobic BS pretreatment of tobacco straw, the following EMYs were found: 251.7, 349.1, 294.6, 282.7, and 301.1 mL/g VS at oxygen concentrations of 0, 4, 8, 12, and 16 mL/g VS, respectively. BS, as a microbial pretreatment agent, resulted in better methane production in the microaerobic environment than under anaerobic conditions ($^*P < 0.05$). The BS-4-pretreated tobacco straw showed the highest cumulative methane yield with an increase of 19.8 % relative to that (291.5 mL/g VS) of untreated straw ($^*P < 0.05$). With increasing oxygen concentration, the methane yield of pretreated tobacco straw decreased compared to that of the BS-4 group ($^*P < 0.05$). This might have occurred because the key facultative hydrolysis bacteria in BS, such as *Enterococcus,* tend to have good performance under low oxygen partial pressures (Tholen et al., 1997).

Two-way analysis of variance was also carried out to investigate the effects of pretreatment types and oxygen concentrations on methane production of tobacco straw. The result showed that pretreatment agent and oxygen concentration were both the important factors to influence the methane production ($^*P < 0.01$), and there was a significant interaction between these two factors ($^*P < 0.01$). This result suggested that the effect of microaerobic pretreatment on AD was closely dependent on the microbial agent and oxygen concentration. Overall, it could be

Table 1
Characteristics of substrates and microbial agents.

Parameters	Tobacco straw	BS	SI	LI
VS (%)[a]	6.8 ± 0.12	4.7 ± 0.27	58.3 ± 0.65	1.7 ± 0.64
Ash (%)[a]	1.1 ± 0.02	2.8 ± 0.20	39.8 ± 1.25	1.1 ± 0.71
VS/TS (%)	86.0 ± 0.65	62.7 ± 1.55	59.4 ± 1.76	60.7 ± 0.37
C (%)[b]	42.4 ± 0.21	30.5 ± 0.64	NA	NA
H (%)[b]	5.8 ± 0.02	4.3 ± 0.31	NA	NA
O (%)[b]	49.6 ± 0.24	62.2 ± 0.35	NA	NA
N (%)[b]	1.7 ± 0.07	2.5 ± 0.04	NA	NA
S (%)[b]	0.2 ± 0.01	0.4 ± 0.02	NA	NA
Cellulose (%)[b]	23.08	NA	NA	NA
Hemicellulose (%)[b]	23.48	NA	NA	NA
Lignin (%)[b]	19.68	NA	NA	NA

NA: Not available; [a] On the standard of the total weight; [b] On the standard of the TS.

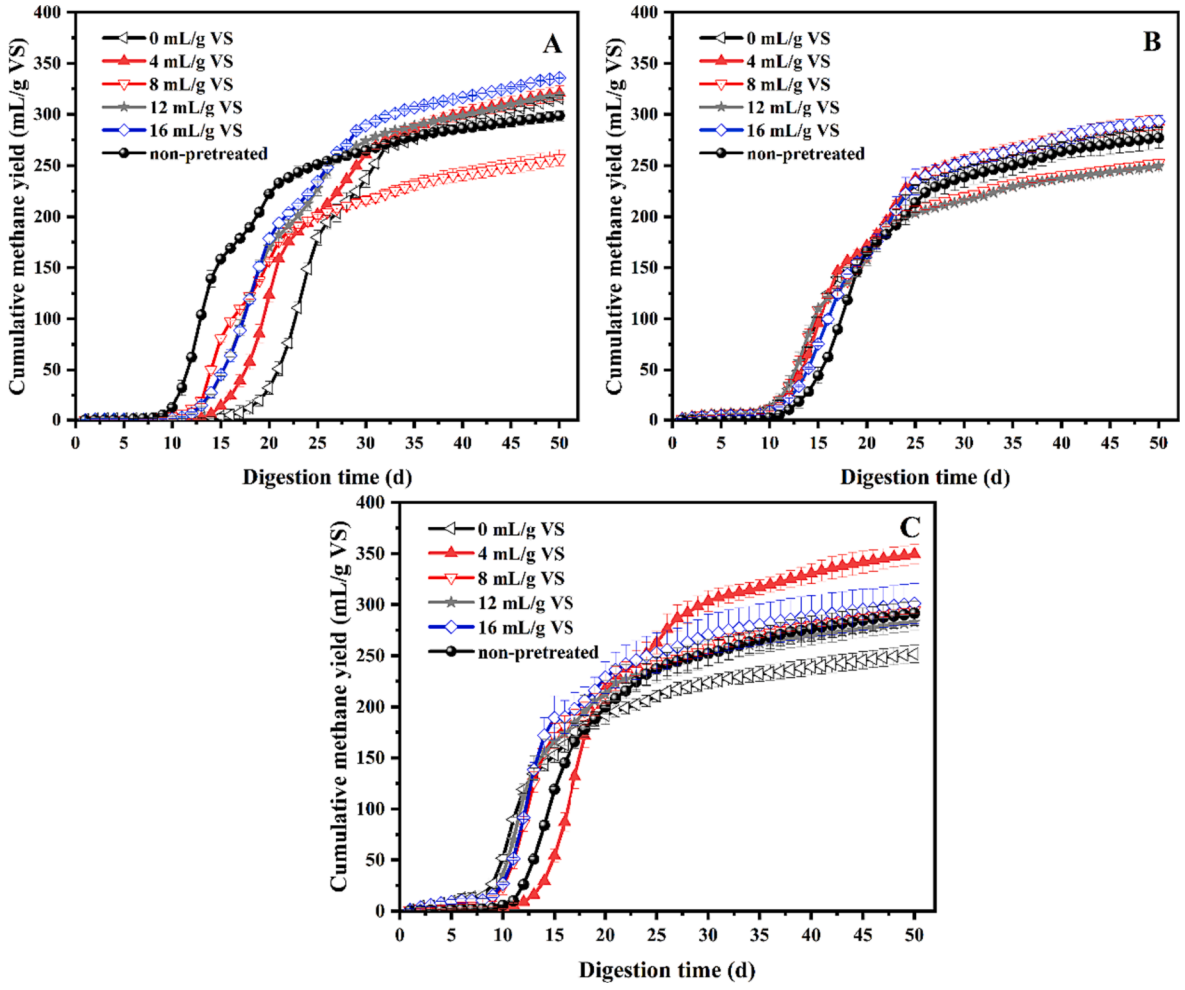


Fig. 1. Cumulative methane yield from AD of tobacco straw after various microaerobic pretreatments; A: (SI), B: (LI), C: (BS).

concluded that LI-16, SI-16, and BS-4 had the highest methane production during the respective pretreatment groups. Among them, the methane production of BS-4 group was significantly higher than SI-16 and LI-16 group ($^*P < 0.05$). These indicated that BS as a microbial pretreatment agent under optimal oxygen conditions was likely to be an effective strategy for enhancing the methane production of tobacco straw.

3.2. *Volatile fatty acids (VFAs) and lactic acid fluctuation after different microaerobic pretreatments*

The variation in the VFAs and lactic acid content due to microaerobic pretreatment to some extent reflects whether this technology could increase hydrolysis to yield more small molecular soluble substances for acid production (Eryildiz et al., 2020; Valentino et al., 2021). As seen in Fig. 2, microaerobic pretreatment with the SI and BS agents resulted in higher VFAs and lactic acid production than that under anaerobic condition, while only the LI agent at an oxygen concentration of 4 mL/g VS promoted acid production. Overall, this phenomenon might suggest that microaerobic conditions could enhance hydrolysis efficiency to promote acid accumulation. In the SI pretreatment group, acetate predominated at a concentration of 2640.0 mg/L in the reactor after anaerobic pretreatment, and its content decreased less after oxygen was injected into the pretreatment system. Propionate could be a valid indicator for the potential performance of tobacco straw hydrolysis in the pretreatment phase (Ohnmacht et al., 2022), and its concentrations in all microaerobic reactors increased. The SI-16 group resulted in the highest propionate yield (1349.2 mg/L), suggesting that this condition possibly had a better hydrolysis performance. Lactic acid is an important indicator of AD stability (Cheng et al., 2020), and high concentrations of lactic acid tend to hinder methanogenic microorganism activity (Balasundaram et al., 2022). The average lactic acid content detected in the reactors with various SI pretreatments followed the order of 1900.0 mg/L (SI-0) > 1700.0 mg/L (SI-8) > 1650.0 mg/L (SI-4, 12) > 1450.0 mg/L (SI-16). Similarly, the lactic acid content decreased in the reactor with various LI pretreatments along with the progressive increase in oxygen concentration. The content of lactic acid was relatively lower in the SI-16 and LI-16 groups, which might mean that the methane production process was less hindered, and this could be one reason why SI-16 and LI-16 had the highest methane production in their groups.

Fig. 2 indicates that the acetate content in BS-0, 4, 8, 12, and 16 were 458.2, 2386.6, 1353.0, 2292.8, and 1172.2 mg/L, respectively, and the highest value of 2386.6 mg/L was achieved when 4 mL/g VS oxygen was introduced in the BS pretreatment process. With BS-4 application, relatively higher production of both propionate (739.8 mg/L) and butyrate (6412.6 mg/L) was observed in relation to other conditions. The high butyrate content observed might be related to butyric acid-producing microorganisms such as *Clostridium*, but this requires further confirmation. On the other hand, no lactic acid was detected in the BS pretreatment system regardless of whether oxygen was present or absent. It was possible that the microorganisms in the BS possess the ability to convert lactic acid into acetic acid or that the rate of lactic acid production did not match its conversion rate. It could be seen that BS-4 pretreatment had the ability to produce more intermediate acids for

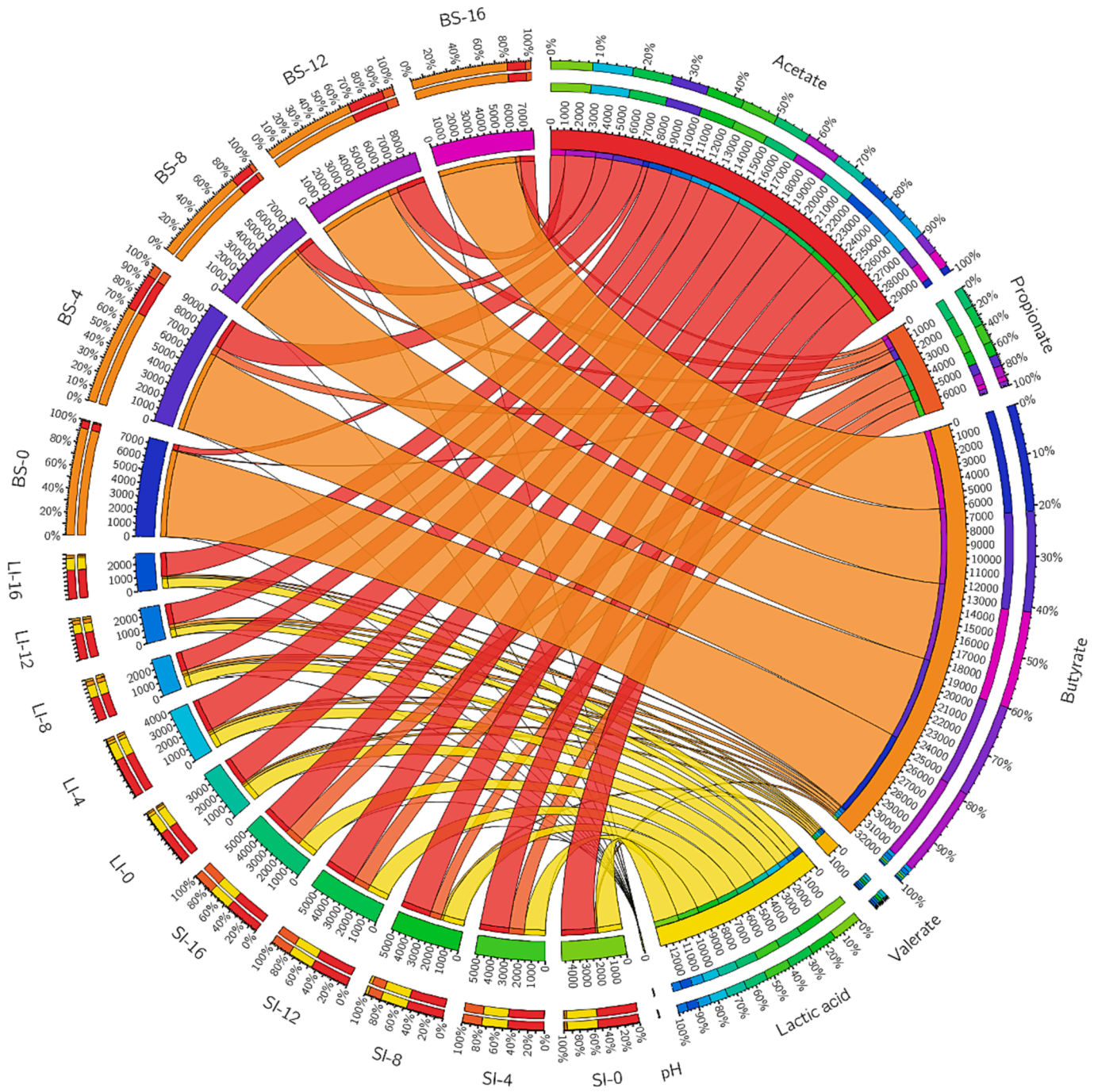


Fig. 2. Volatile fatty acids (VFAs) and lactic acid contents after various microaerobic pretreatments. Inner circle: acid concentration (mg/L); Outer circle: acid percentage (%).

methanogenesis, and this could be the reason for the predominance in methane production of tobacco straw under this condition.

3.3. Microstructural analysis

Microbial pretreatments via the injection of optimal concentrations of oxygen (SI-16, LI-16, BS-4) may influence the physicochemical and structural properties of tobacco straw; therefore, FTIR and XRD monitoring were performed to observe structure changes in this study (see supplementary material).

Infrared absorption spectra, one indicator of functional groups, could be applied to discover the relative digestibility of the lignocellulose in tobacco straw. Tobacco straw with and without optimal microaerobic pretreatment was quite different, indicating major differences and overlap in the spectral regions assigned to the tensile vibrations of –OH in lignocellulose (3323.0 cm^{-1}), stretching vibrations of the CH_3-CO– group in hemicellulose (1734.0 cm^{-1}), tensile and skeletal vibrations of C═C in the aromatic ring of lignin (1626.0 cm^{-1}, 1510.0 cm^{-1}), asymmetric stretching vibration of C-O-C in cellulose and C-O stretching vibrations of cellulose and hemicellulose (1155.0 cm^{-1}, 1054.0 cm^{-1}) (Kumar et al., 2019; Taherdanak and Zilouei, 2014). The transmittance rates at 3323.0–3418.0 cm^{-1}, 1734.0 cm^{-1}, 1155.0 cm^{-1}, and 1054.0 cm^{-1} were in the order BS-4 > SI-16 > LI-16 > untreated; the transmittance rates at 1626.0 and 1510.0 cm^{-1} were in the order SI-16 > LI-

16 > BS-4 > untreated. In general, the lower the transmittance band is, the stronger the absorbance. In other words, the higher transmittance bands at 3323.0–3418.0 cm^{-1}, 1734.0 cm^{-1}, 1155.0 cm^{-1}, and 1054.0 cm^{-1} of the BS-4-pretreated tobacco straw meant that the weaker absorbance for the cellulose and hemicellulose, suggesting a stronger decomposition in terms of cellulose and hemicellulose. Accordingly, the transmittance band from BS-4 pretreated tobacco straw assigned to lignin was relatively lower, and thus, the relative absorbance attributed to lignin was higher than that from the other pretreatment conditions, which might be due to the relative content of lignin increased as the decomposition of cellulose and hemicellulose. The introduction of 4 mL/g VS in the BS pretreatment process might have a better hydrolysis performance for cellulose and hemicellulose, and therefore resulted in the highest methane production in this study.

The identification and quantification of the crystallinity and amorphous phase in tobacco straw were performed utilizing XRD (see supplementary material). Given the XRD data, the calculated order of cellulose crystallinity was BS-4 (29.9 %) > SI-16 (28.6 %) > untreated (17.3 %) > LI-16 (17.1 %), among which untreated and LI-16 tobacco straw were insignificantly different in the *CrI*. BS-4 and SI-16 contributed to the strengthening of *CrI*, exhibiting a higher relative content of cellulose primarily in the crystal state. The underlying reason for this might be that cellulose, hemicellulose, and lignin in the amorphous region were significantly decomposed, which resulted in an increase in the relative content of cellulose in the crystalline form.

3.4. Kinetic analysis

A modified Gompertz model was utilized to serve as an important reference for understanding the different AD processes in this study. The kinetic curve and fitted parameters of all the reactions were computed by a statistical algorithm, and model sensitivity was analysed compare the experimental results and the simulated values; the fit was considered good at an R^2 value over 0.98 as shown in Table 2. λ is associated with the lag period of the process, and its values for BS-4, SI-16, and LI-16 are 12.3, 12.8, and 10.9, respectively. In comparison, the lag period of BS-4 did not significantly differ from that of SI-16 and was higher than that of LI-16. However, the gap between BS-4 and LI-16 in the lag period might have been compensated by the higher methane production in the BS-4 group. For the maximum methane production, all Y_0 results are significantly close to the experimental methane production. BS-4 pretreatment showed the most effective lignocellulose decomposition, as it had the highest *BD* value of 61.9 %, followed by 59.6 % in BS-16 pretreated straw, further showing good agreement with the methane production analysis above.

3.5. Microbial analysis

3.5.1. Bacterial composition at the genus level

The bacterial composition of the three microbial agents as well as that in the reactors prior and after optimal pretreatment and AD process is shown at the genus level in Fig. 3. The top 2 bacterial genera with high abundance in SI were *Bacillus* and *Streptomyces*, with relative abundances of 60.6 % and 13.7 %, respectively. *Bacillus*, a strictly aerobic or facultative anaerobic bacterium affiliated with *Firmicutes*, is responsible for the hydrolysis of large lignocellulosic molecules into soluble sugars (Wang et al., 2021; Zabed et al., 2019), and its relative abundances were 3.0 % and 0.9 % after SI-0 and SI-16 pretreatment, respectively, and then changed to negligible level at the end of AD. The FTIR results indicated that the peaks associated with cellulose, hemicellulose, and lignin all decreased in pretreated tobacco straw with SI compared with untreated tobacco straw. The reason behind this result might be that *Bacillus* could secrete a variety of enzymes (e.g., cellulase, xylanase, amylase, peroxidase) and thus promote lignocellulose dissolution during pretreatment stage. *Streptomyces*, a bacterium belonging to the phylum *Actinomycetes*, is widely present in soils and plays an important role in the degradation of organic matter (Hong et al., 2010). The relative abundances of *Streptomyces* were 0.3 % and 0.1 % after SI-0 and SI-16 pretreatment with insignificant difference between them, and after AD, this bacterium was hardly detected. It could secrete a lytic polysaccharide monooxygenase, which was initially found to degrade chitin and later was shown to destroy the glycosidic bond of cellulose. Lytic polysaccharide monooxygenase primarily decomposes refractory polysaccharides in the crystalline zone of lignocellulose through oxidation and results in the generation of oxidized oligosaccharides (Hong et al., 2010). Additionally, this enzyme could work with hydrolase to increase the hydrolysis efficiency of lignocellulose (Eibinger et al., 2014; Sharma et al., 2019), which might promote the degradation of lignocellulose. At the same time, there is significant increases in the relative abundance of *Lactobacillus* from 2 % in the SI to 30.6 % and 60.0 % in the SI-0 and SI-16 pretreatment processes, respectively. At the end of AD, the relative abundance of *Lactobacillus* was reduced to a negligible level. *Lactobacillus* belong to the family *Lactobacillaceae*, among which anaerobic, facultative, and aerobic microorganisms could play an important role in

Table 2
Biodegradability and kinetic parameters on AD of tobacco straw.

Pretreatment	Oxygen concentration (mL/g VS)	CMY (mL/g VS)	*BD* (%)	Y_0 (mL/g VS)	μ_m (mL/g VS d)	λ (d)	R^2
SI	0	315.9 ± 3.14	56.1	303.7	25.2	18.6	0.998
	4	321.1 ± 6.89	57.0	311.2	21.2	14.8	0.996
	8	257.7 ± 7.74	45.7	243.8	17.6	11.1	0.994
	12	318.2 ± 3.37	56.5	307.1	21.3	12.7	0.997
	16	335.8 ± 0.4	59.6	325.5	22.1	12.8	0.998
	untreated	298.7 ± 4.93	53.0	284.4	22.5	8.9	0.992
LI	0	286.3 ± 0.41	50.8	277.3	17.4	9.8	0.996
	4	292.6 ± 7.35	51.9	282.4	18.5	10.2	0.996
	8	252.4 ± 3.06	44.8	243.0	16.3	9.4	0.995
	12	249.2 ± 0.09	44.2	239.9	15.6	9.1	0.995
	16	293.3 ± 5.41	52.0	282.7	18.7	10.9	0.997
	untreated	277.3 ± 10.38	49.2	265.6	20.1	12.4	0.996
BS	0	251.7 ± 8.58	44.7	237.7	16.9	6.2	0.988
	4	349.1 ± 9.46	61.9	333.1	26.0	12.3	0.994
	8	294.6 ± 2.72	52.3	278.0	20.4	7.8	0.987
	12	282.7 ± 7.95	50.2	267.8	20.1	7.3	0.989
	16	301.1 ± 19.68	53.4	285.3	22.7	7.8	0.987
	untreated	291.5 ± 11.98	51.7	275.4	21.7	10.2	0.992

CMY: Cumulative methane yield.

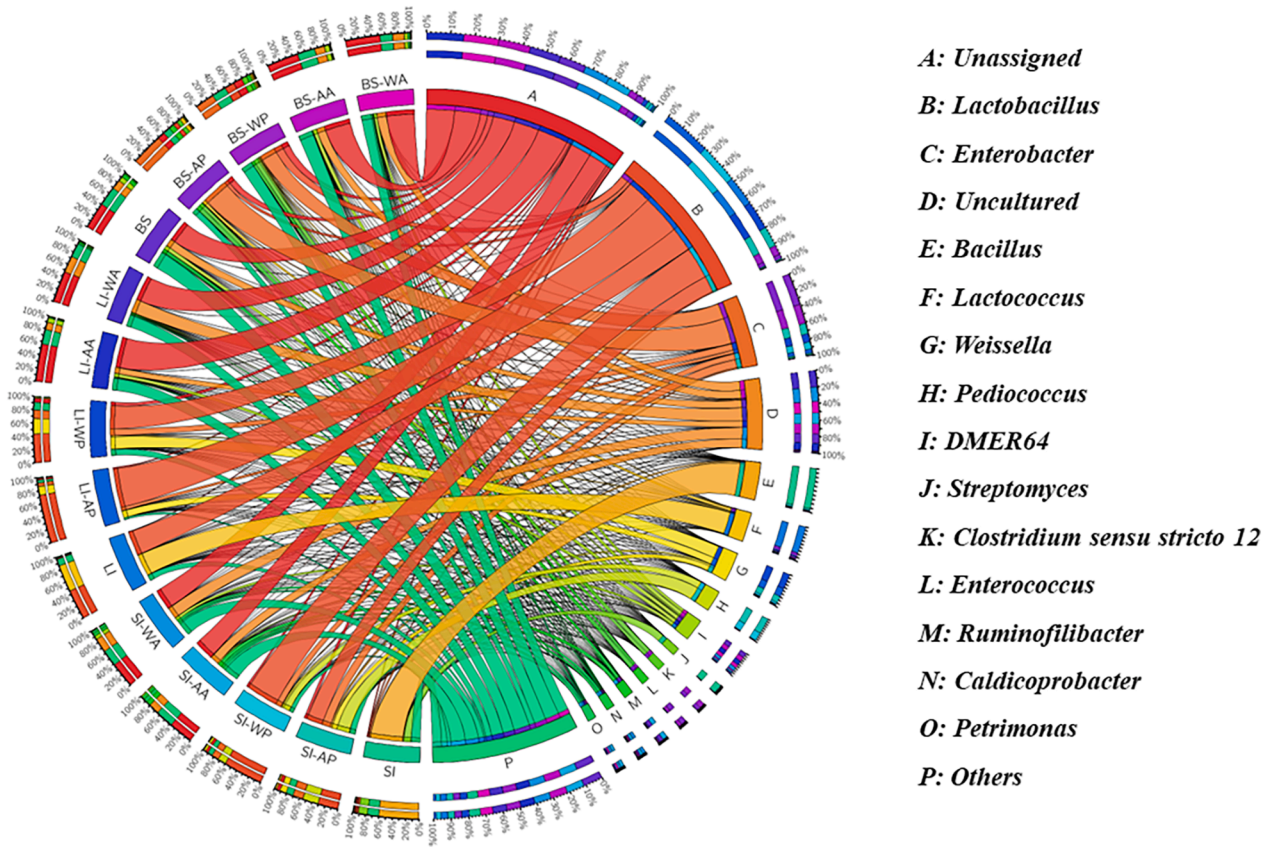


Fig. 3. Bacterial relative abundance on the genus level. AP: anaerobic pretreatment; MP: optimal microaerobic pretreatment; AA: anaerobic digestion after anaerobic pretreatment; MA: anaerobic digestion after optimal microaerobic pretreatment.

the hydrolysis and acidification process (D' Silva et al., 2021), with functions involving the fermentation of sugar to release acetic and lactic acid for bacterial development and subsequent methanogenesis. Therefore, the higher relative abundance of *Lactobacillus* in SI-16 pretreatment process was probably resulted from the increasing facultative microorganisms in the genus *Lactobacillus*, and this was consistent with the result of Nguyen et al. (2019) who also found microaerobic condition was good for the enrichment of facultative bacterium. Based on the microbial community analysis, optimal oxygen injection in SI pretreatment process could significantly enhance the relative abundance of facultative bacterium in genus *Lactobacillus*, and then produce more acetic and lactic acid, which was likely to be the explanation for the increase in methane production with SI-16 pretreatment application.

Among the LI agents, *Lactobacillus* and *Lactococcus* were the top 2 bacteria, with relative abundances of 45.3 % and 41.1 %, respectively. Both genera belong to *Firmicutes* and are responsible for lactic acid production with sugars as the carbon nutrients (McAteer et al., 2020; Wang et al., 2021). The relative abundances of *Lactobacillus* were 73.5 % and 43.5 % after LI-0 and LI-16 pretreatment, respectively. A possible reason for this phenomenon was that anaerobic microorganisms in the genus *Lactobacillus* in the LI agent treatment were likely to be predominant. The excessive higher abundance of this bacterium in anaerobic pretreatment environment might induce acid accumulation, and therefore had a negative effect on methane production, which could be the reason why LI-16 had higher methane production than that of LI-0. *Weissella* was not the main genus (0.4 %) in LI and accounted for higher proportions at the end of LI-0 (12.8 %) and LI-16 pretreatment (23.0 %), and it was hard to be detected in AD stage. *Weissella* is a facultative anaerobe that belongs to *Firmicutes*, which could digest carbohydrate through the hexose-monophosphate and the phosphoketolase pathways to produce acetate and ethanol (Fusco et al., 2015). Additionally, *Enterobacter* was enriched to 12.7 % in the LI-16 pretreatment process relative to that of 5.8 % in LI-0 condition, which was reported to have special ligninolytic species such as *Enterobacter hormaechei KA3* (Zhang et al., 2021). Upon the optimal dosing of oxygen for the LI pretreatment process, the digestion of carbohydrate in tobacco straw was facilitated via abovementioned pathway from *Weissella*. At the same time, the increase in the relative abundance of *Enterobacter* might play an important role in lignin degradation, and this result was consistent with the FTIR analysis, which suggested that LI-16 had a weaker absorption strength for lignin than untreated. As a whole, optimal oxygen loaded in LI pretreatment process could help the fermentation of carbohydrate and lignin, and therefore enhance the methane production.

In terms of BS, *Lactobacillus* in BS increased from 0.25 % to 6.4 % and 14.8 % in the BS-0 and BS-4 pretreatment environments, respectively. As mentioned above, *Lactobacillus* in BS-4 pretreatment process was identified as a functional microorganism for yielding acetic and lactic acid, and in contrast with the SI and LI groups, the significantly lower average abundance of *Lactobacillus* might have resulted in less lactic acid production in pretreatment process. *Clostridium sensu stricto 12* showed a higher relative abundance of 8.1 % in BS-4 pretreatment process than that of 6.1 % under BS-0 condition, and it is a common bacterium that carries out hydrolysis to produce acetic and butyric acid (Zhao et al., 2022). Interestingly, the pH under this microaerobic environment was approximately 6, showing good agreement with the findings of Zhao et al. (2021) who found that *Clostridium sensu stricto 12* was capable of hydrolysing organic substances into butyric acid at pH = 6. This finding was consistent with the proposed reason for the high butyrate production, as mentioned in the VFAs analysis. *Enterococcus*, a facultative genus, was enriched in both the anaerobic and microaerobic pretreatment phases. Tholen et al. (1997) found that *Enterococcus* could increase

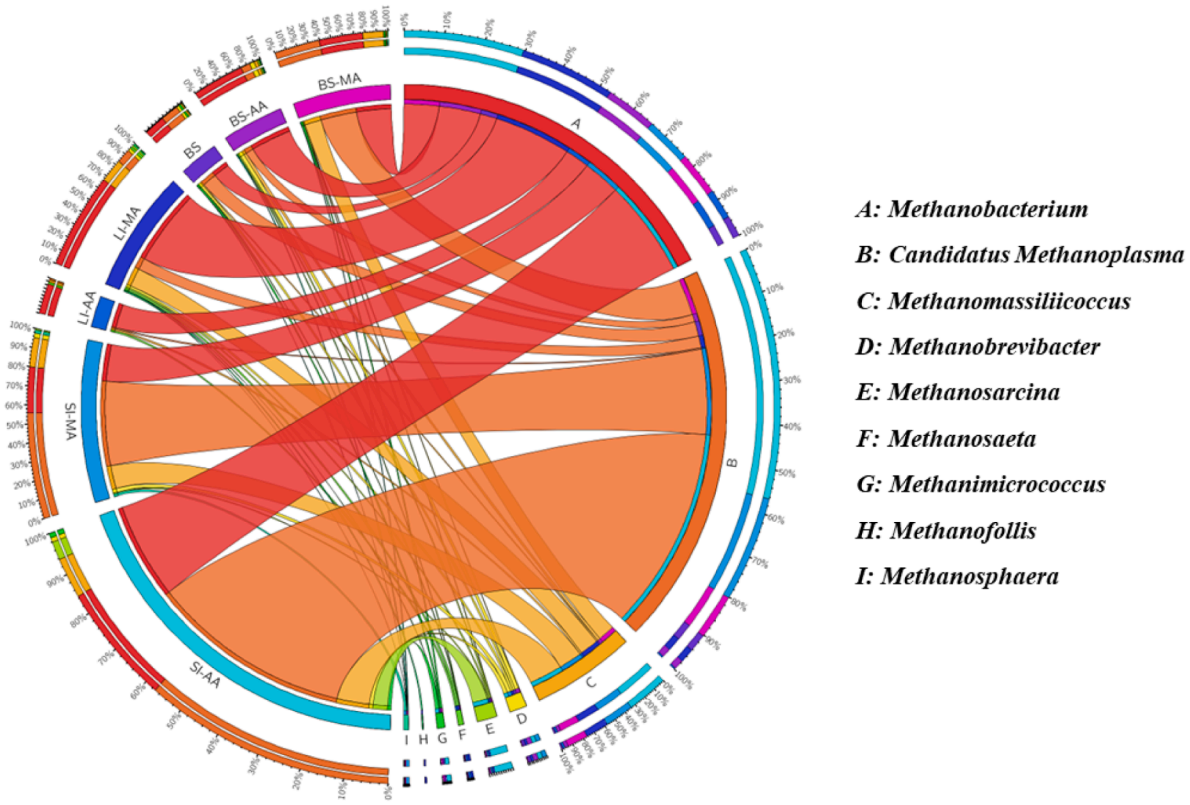


Fig. 4. Archaeal relative abundance on the genus level. AP: anaerobic pretreatment; MP: optimal microaerobic pretreatment; AA: anaerobic digestion after anaerobic pretreatment; MA: anaerobic digestion after optimal microaerobic pretreatment.

the transformation of lactic acid into acetic acid under relatively lower oxygen pressure. Hence, the BS microaerobic pretreatment system might have two characteristics: the first was to rapidly consume the accumulated lactic acid to produce acetic acid under relative lower oxygen concentration, and the second was to increase the relative abundance of genus *Clostridium sensu stricto 12* for enhancing hydrolysis rate. This might be a reason that the lowest oxygen concentration of 4 mL/g VS under pretreatment process was better to increase the methane production when BS was used as microbial agent.

3.5.2. Archaeal composition at the genus level

The archaeal composition analysis at the genus level revealed that *Methnanobacterium*, *Candidatus Methanoplasma*, *Metanomassilicoccus*, *Metanobrevibacter*, *Methanosarcina*, and *Methanosaeta* were present at the end of AD (Fig. 4). Among them, in addition to unassigned archaea, *Methanobacterium* and *Candidatus Methanoplasma* were the most abundant genera. *Methanobacterium*, in the family *Methanobacteriaceae*, is involved in the transformation of carbon dioxide and hydrogen into methane (Amin et al., 2021a). As for SI pretreatment group, *Methanobacterium* was more abundant in the SI-0 (3.3 %) than that of SI-16 (1.3 %), suggesting a dominated hydrogenotrophic methanogenesis might occur on anaerobic environment. In terms of LI pretreatment group, the relative abundances of *Methanobacterium* from LI-0 and LI-16 were 0.8 % and 2.6 %, respectively. This phenomenon was opposed to that in SI group, indicating that oxygen might promote hydrogenotrophic methanogenesis in this system. At the end of AD of BS group, there is no significant difference between BS-0 (1.4 %) and BS-4 (1.2 %) groups. These results showed that the change of archaeal genus might be varied with the pretreatment agent and oxygen concentration, which showed a good accordance with the previous two-way analysis result. *Candidatus Methanoplasma* is believed to use methyl compounds such as methanol, hydrogen, and carbon dioxide for methane synthesis (Vendruscolo et al., 2020). The relative abundances of SI-0 and SI-16 were 7.0 % and 3.0 %, respectively, and *Candidatus Methanoplasma* was rarely found in the AD process with LI as the microbial agent. A higher abundance (1.3 %) of the archaeal genus *Candidatus Methanoplasma* was observed in BS-16 than in BS-0 (0.2 %), and this indicated that the presence of oxygen in pretreatment step using BS might be good for methanogenesis process.

4. Conclusions

The pretreatment of tobacco straw via BS application as a microbial agent at an oxygen concentration of 4 mL/g VS resulted in the greatest increase in VFAs and methane production. This might occur due to the high relative abundance of *Enterococcus* and *Clostridium sensu stricto 12* in the pretreatment phase, which led to better hydrolysation of lignocellulose. In AD, *Candidatus Methanoplasma,* which can transform most of methyl compounds into methane, might also be a reason for the superior methane production under this condition. This study advances the understanding of microaerobic pretreatment and offers valuable guidelines to effectively utilize tobacco straw.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (52270116).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2023.130092.

References

Amin, F.R., Khalid, H., El-Mashad, H.M., Chen, C., Liu, G., Zhang, R., 2021a. Functions of bacteria and archaea participating in the bioconversion of organic waste for methane production. Sci. Total Environ. 763, 143007.

Amin, F.R., Khalid, H., Li, W., Chen, C., Liu, G., 2021b. Enhanced methane production and energy potential from rice straw by employing microaerobic pretreatment via anaerobic digestion. J. Clean. Prod. 296, 126434.

APHA, 2012. Standard methods for the examination for water and wastewater, twenty-third ed. American Public Health Association, Washington DC, USA.

Balasundaram, G., Banu, R., Varjani, S., Kazmi, A.A., Tyagi, V.K., 2022. Recalcitrant compounds formation, their toxicity, and mitigation: Key issues in biomass pretreatment and anaerobic digestion. Chemosphere 291, 132930.

Buswell, A., Mueller, H., 1952. Mechanism of methane fermentation. Ind. Eng. Chem. 44 (3), 550–552.

Chen, Y., Ping, Q., Li, D., Dai, X., Li, Y., 2022. Comprehensive insights into the impact of pretreatment on anaerobic digestion of waste active sludge from perspectives of organic matter composition, thermodynamics, and multiomics. Water Res. 226, 119240.

Cheng, J., Hua, J., Kang, T., Meng, B., Yue, L., Dong, H., Li, H., Zhou, J., 2020. Nanoscale zero-valent iron improved lactic acid degradation to produce methane through anaerobic digestion. Bioresour. Technol. 317, 124013.

D' Silva, T.C., Isha, A., Chandra, R., Vijay, V.K., Subbarao, P.M.V., Kumar, R., Chaudhary, V.P., Singh, H., Khan, A.A., Tyagi, V.K., Kovács, K.L., 2021. Enhancing methane production in anaerobic digestion through hydrogen assisted pathways – A state-of-the-art review. Renew. Sust. Energ. Rev. 151, 111536.

Díaz, I., Donoso-Bravo, A., Fdz-Polanco, M., 2011. Effect of microaerobic conditions on the degradation kinetics of cellulose. Bioresour. Technol. 102 (21), 10139–10142.

Eibinger, M., Ganner, T., Bubner, P., Rošker, S., Kracher, D., Haltrich, D., Ludwig, R., Plank, H., Nidetzky, B., 2014. Cellulose surface degradation by a lytic polysaccharide monooxygenase and its effect on cellulase hydrolytic efficiency. J. Biol. Chem. 289 (52), 35929–35938.

Eryildiz, B., Lukitawesa, Taherzadeh, M.J., 2020. Effect of pH, substrate loading, oxygen, and methanogens inhibitors on volatile fatty acid (VFA) production from citrus waste by anaerobic digestion. Bioresour. Technol. 302, 122800.

Fusco, V., Quero, G.M., Cho, G.S., Kabisch, J., Meske, D., Neve, H., Bockelmann, W., Franz, C.M., 2015. The genus *Weissella*: taxonomy, ecology and biotechnological potential. Front. Microbiol. 6, 155.

Hong, Z., Morten, S., Svein, J., Horn, B., Westereng, G., 2010. An oxidative enzyme boosting the enzymatic conversion of recalcitrant polysaccharides. Science 330(Oct. 8 TN.6001), 219–222.

Huiliñir, C., Pagés-Díaz, J., Vargas, G., Vega, S., Lauzurique, Y., Palominos, N., 2023. Microaerobic condition as pretreatment for improving anaerobic digestion: A review. Bioresour. Technol. 384, 129249.

Kumar, S., Gandhi, P., Yadav, M., Paritosh, K., Pareek, N., Vivekanand, V., 2019. Weak alkaline treatment of wheat and pearl millet straw for enhanced biogas production and its economic analysis. Renew Energ. 139, 753–764.

Lim, J.W., Wang, J.Y., 2013. Enhanced hydrolysis and methane yield by applying microaeration pretreatment to the anaerobic codigestion of brown water and food waste. Waste Manag. 33 (4), 813–819.

McAteer, P.G., Christine Trego, A., Thorn, C., Mahony, T., Abram, F., O'Flaherty, V., 2020. Reactor configuration influences microbial community structure during high-rate, low-temperature anaerobic treatment of dairy wastewater. Bioresour. Technol. 307, 123221.

Nguyen, D., Wu, Z., Shrestha, S., Lee, P., Raskin, L., Khanal, S., 2019. Intermittent micro-aeration: New strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res. 166, 115080.

Ohnmacht, B., Lemmer, A., Kress, P., Steinbrenner, J., Oechsner, H., 2022. Propionic acid degradation in anaerobic digestion and recovery strategies for full-scale biogas plants. Bioresour. Technol. Rep. 19, 101192.

Okuda, N., Ninomiya, K., Takao, M., Katakura, Y., Shioya, S., 2007. Microaeration enhances productivity of bioethanol from hydrolysate of waste house wood using ethanologenic *Escherichia coli* KO11. J. Biosci. Bioeng. 103 (4), 350–357.

Sharma, H.K., Xu, C., Qin, W., 2019. Biological pretreatment of lignocellulosic biomass for biofuels and bioproducts: an overview. Waste Biomass Valori. 10, 235–251.

Soest, V., Robertson, J.B., Lewis, B.A., 1991. Polysaccharides in relation to animal nutrition. J. Dairy Sci. 74, 3583e3597.

Surendra, K.C., Ogoshi, R., Zaleski, H.M., Hashimoto, A.G., Khanal, S.K., 2018. High yielding tropical energy crops for bioenergy production: Effects of plant components, harvest years and locations on biomass composition. Bioresour. Technol. 251, 218–229.

Taherdanak, M., Zilouei, H., 2014. Improving biogas production from wheat plant using alkaline pretreatment. Fuel 115, 714–719.

Tholen, A., Schink, B., Brune, A., 1997. The gut microflora of *Reticulitermes flavipes*, its relation to oxygen, and evidence for oxygen-dependent acetogenesis by the most abundant *Enterococcus sp*. FEMS Microbiol. Ecol. 24 (2), 137–149.

Valentino, F., Munarin, G., Biasiolo, M., Cavinato, C., Bolzonella, D., Pavan, P., 2021. Enhancing volatile fatty acids (VFA) production from food waste in a two-phases pilot-scale anaerobic digestion process. J. Environ. Chem. Eng. 9 (5), 106062.

Vendruscolo, E., Mesa, D., Rissi, D., Meyer, B., de Oliveira Pedrosa, F., de Souza, E., Cruz, L., 2020. Microbial communities network analysis of anaerobic reactors fed with bovine and swine slurry. Sci. Total Environ. 742, 140314.

Wang, L., Liu, C., Wei, B., Song, C., Cai, F., Liu, G., Chen, C., 2021. Effects of different microbial pretreatments on the anaerobic digestion of giant grass under anaerobic and microaerobic conditions. Bioresour. Technol. 337, 125456.

Zabed, H.M., Akter, S., Yun, J., Zhang, G., Awad, F.N., Qi, X., Sahu, J.N., 2019. Recent advances in biological pretreatment of microalgae and lignocellulosic biomass for biofuel production. Renew. Sust. Energ. Rev. 105, 105–128.

Zhang, Q., Zhang, J., Zhao, S., Song, P., Chen, Y., Liu, P., Mao, C., Li, X., 2021. Enhanced biogas production by ligninolytic strain *Enterobacter hormaechei ka3* for anaerobic digestion of corn straw. Energies 14 (1), 2990.

Zhang, X., Zhu, X., Duan, J., Li, H., Xu, L., Xiong, L., Luo, Q., Yang, M., Chang, X., 2023. Comparative study on properties of polypropylene-based composites reinforced with tobacco stalk fibers (unmodified / modified) from different parts of tobacco stalk. Ind. Crop. Prod. 201, 116890.

Zhao, Q., Arhin, S.G., Yang, Z., Liu, H., Wang, W., 2021. pH regulation of the first phase could enhance the energy recovery from two-phase anaerobic digestion of food waste. Water Environ. Res 93 (8), 1370–1380.

Zhao, W., Yan, B., Ren, Z.J., Wang, S., Zhang, Y., Jiang, H., 2022. Highly selective butyric acid production by coupled acidogenesis and ion substitution electrodialysis. Water Res. 226, 119228.

Zhen, F., Luo, X., Xing, T., Sun, Y., Kong, X., Li, W., 2021. Performance evaluation and microbial community analysis of microaerobic pretreatment on thermophilic dry anaerobic digestion. Biochem. Eng. J. 167, 107873.